



Contrasting financial and business cycles: Stylized facts and candidate explanations[☆]



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ABSTRACT

This paper contrasts the empirical features of financial and business cycles of 13 European Union countries, and discusses candidate theoretical mechanisms which could explain these differences. Relative to their business cycle counterparts, we show that financial cycles have a higher amplitude, a longer duration and exhibit far greater symmetry. Mostly owing to this difference in symmetry, we find that financial and business cycles synchronise on average only two-thirds of the time, exhibiting weaker concordance in countries having experienced very persistent financial downturns. We then relate our empirical results to the literature by reviewing five model mechanisms that could potentially underlie these facts. Overall, our results suggest that macroeconomic stabilization and financial stability objectives can, at times, be in conflict.

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1. Introduction

In the wake of the global financial crisis, understanding the interaction of financial and business cycles has become a critical policy issue, given the potential for conflicts in price and financial stability objectives (see, for instance, Smets (2014)). Such conflicts lend support to those arguing for a separation of responsibilities between monetary and macroprudential authorities, consistent with Tinbergen's effective assignment principle.¹ A targeted ability to stabilise national cycles may be particularly powerful in a currency union such as the euro area, which binds countries through a common monetary policy.

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¹ In fact, monetary and macroprudential policies alone may not be a sufficient set of policies for cycle stabilisation. Indeed, the perimeter of policies to address unique cycles could also extend more widely to other macroeconomic policies (such as fiscal and structural) as well as other financial policies.

This study contributes to the debate on reconciling price and financial stability objectives in two ways. First, it characterizes the properties of financial cycles, offering new evidence on their relationship with business cycles *within* 13 European Union countries. Second, we discuss multiple candidate economic mechanisms that might underlie these empirical relationships. By relating the novel stylized facts that we document to the literature, our study sheds light on the following three questions: How do the main characteristics of financial cycles compare with those of business cycles? To what extent are financial and business cycles synchronized? What economic mechanisms could explain the properties of financial cycles and their differences with respect to business cycles?

Empirical results that follow from turning point analysis suggest that financial cycles are distinct from business cycles in several ways. First, financial cycles tend to have a higher amplitude – with credit and house price growth around twice as volatile as GDP growth. Second, financial cycles have a longer duration – with an average length of 11.5 years – and are more than 60% longer than their business cycle counterparts. Third, financial cycle phases exhibit far greater symmetry than business cycles – with persistent financial downturns contrasting with the short duration of recessions, generally lasting only just over a year. As business cycle recessions are short-lived, the longer duration of finan-

cial cycle downturns is the key dimension along which the two cycles are found to diverge. Mostly owing to the more symmetric nature of financial cycles, we find that financial and business cycles synchronize only around two-thirds of the time. Periods of disconnect between financial and real variables are therefore frequently observed – suggesting that financial imbalances could grow undetected in stable macroeconomic environments.

For the European Union countries analysed in this study, several mechanisms could potentially explain the above-mentioned three characteristics of financial cycles. Debt overhang effects are likely to be a contributing factor, as periods of debt build-up give way to lengthy periods of deleveraging. Other competing mechanisms outlined in the theoretical literature could also play a role – including financial shocks embedding accelerator-like properties, expectational errors, and other types of market failures stemming from strategic complementarities or structural features of national housing markets.

The notion that the diverging dynamics of financial and business cycles stems from different types of underlying frictions chimes well with recent literature. As shown by Collard et al. (2017), if such frictions take the form of excessive risk-taking and nominal rigidities, the presence of two distinct market failures implies a clear-cut division of tasks between monetary and prudential policies. Similarly, Carillo et al. (2017) show that violations of the Tinbergen principle, as well as strategic interactions between the different authorities, can significantly undermine the effectiveness of both monetary and prudential policies.² The distinct features of financial cycles that we document could therefore imply a need for a targeted and differentiated policy with respect to such targeting business cycles. In other words, separate policy instruments may be needed to reconcile potentially conflicting financial and economic stability objectives.

The empirical findings of this study relate to a rapidly emerging literature analysing the properties of financial variables, such as credit and asset prices, in relation to business cycles (see Claessens et al. (2011, 2012), Drehmann et al. (2012); Hiebert et al. (2014); Aikman et al. (2015); Schuler et al. (2015, 2017); Strohsal et al. (2015a, b); Rünstler and Vlekke (2016); Galati et al. (2016); Verona (2016); Schuler (2018)). Specifically, our study is closely related to the seminal studies by Drehmann et al. (2012) and Claessens et al. (2011, 2012), who find that financial cycles are longer and larger than business cycles and that recessions are more severe when they coincide with financial downturns. We extend their analyses by explicitly studying the concordance of financial and business cycles and their asymmetry using the standard version of the turning point algorithm.³ While Claessens et al. (2011, 2012) analyse single variables, Drehmann et al. (2012), similarly to our study, extract a reference cycle from a set of financial variables (credit, credit-to-GDP, property prices).⁴ Our identification differs from the latter study in that we do not adjust the turning point algorithm to consider only cycles lasting longer than five years for

capturing a financial cycle. Instead, we use the classical version – commonly applied in the business cycle context – that requires cycles to last at least five quarters. Even though Drehmann et al. (2012) have provided a first and important step towards characterising financial cycles, given our purpose, it is important to use the standard rules. Applying the classical version of the algorithm allows us to consistently contrast financial cycles with business cycles. Furthermore, it also allows us to benchmark results to the business cycle literature (see, for instance, Bry and Boschan (1971); Watson (1994); Harding and Pagan (2002); Jordà et al. (2017)). Our application of the classical version of the turning point algorithm also contrasts with Claessens et al. (2011, 2012), who restrict their focus to recovery periods, whereas we consider downturns as well as upturns to study complete cycles. Identifying complete cycles allows us to analyse financial cycle and business cycle concordance and asymmetry.

Our approach builds on Schuler et al. (2017). For G7 countries, the authors show that reference financial cycles based on the common turning points in credit and house prices are consistent in duration with those from other widely used methods (spectral analyses, structural time series analyses, and wavelet techniques).⁵ We extend Schuler et al. (2017) with a systematic review of the concordance of derived financial and business cycles and their symmetry for European Union countries, i.e., in a single market which *a priori* would be expected to have a relatively high degree of convergence. In so doing, we not only present new findings to the literature but also strengthen the basis for policy design through documenting key stylized facts, such as amplitude, duration, and asymmetry.

The remainder of the paper is organised as follows. Section 2 outlines the reduced form turning point analysis. Section 3 discusses candidate economic mechanisms potentially underlying the empirical facts. Section 4 concludes.

2. What can turning point analysis tell us about the similarities and differences between financial and business cycles?

We identify financial cycles through constructing a reference cycle that dates the common peaks and troughs of total credit and house prices. For this, we rely on the widely used turning point algorithm. It has been applied to financial cycles by Schuler et al. (2017), who show that key properties such as the length of financial cycles obtained using this turning point algorithm are broadly equivalent to those obtained by other approaches (e.g., spectral analyses or unobserved components models).

Credit and house prices are closely related for a broad range of economies. Empirically, this is found by Galati et al. (2016), Rünstler and Vlekke (2016), and Schuler et al. (2017). House prices in relation to credit can distinguish good from bad credit booms – that is, those that lead to financial crises (see Mendoza and Terrones (2008) and Gorton and Ordoñez (2016)). Indeed, all major crises in advanced economies since the mid-1970s have been associated with housing busts (see Reinhart and Rogoff (2009)). Furthermore, the relation of credit and house prices is closely linked to the concept of leverage cycles (see Geanakoplos (2010)) and has found its way into popular DSGE models, such as Iacoviello (2005), who introduces

² Van der Ghote (2017) also shows that coordination between monetary and macroprudential policies helps reduce the risk of financial crises.

³ Drehmann et al. (2012) already mention that contractionary phases of financial cycles are longer than typical business cycle downturns. However, they do not provide detailed results on the duration of individual financial cycle phases. They only offer figures of the overall duration of their (reference) financial cycles (see Table 8). Further, these figures are obtained using the non-standard version of the turning point algorithm and, thus, make a comparison with typical business cycle facts difficult.

⁴ In our analysis, we omit credit-to-GDP as an indicator since studies have shown that credit and GDP are governed by different cyclical patterns (see, for instance, Schuler et al. (2015, 2017) or Rünstler and Vlekke (2016)). Further, we would like to study the relation of financial cycles and business cycles measured by GDP. For this reason, a mixture of financial and business cycle indicators would impede the analysis.

⁵ Studies that use the turning point algorithm on individual financial variables do not find a large difference from business cycles (see, for instance, Claessens et al. (2011, 2012) or Hiebert et al. (2014)), which contrasts with other methods which find that financial cycles have a longer duration. Considering the joint fluctuations credit and asset prices in order to determine financial cycles is advanced in several studies (e.g., Drehmann et al. (2012) or Schuler et al. (2015, 2017)). A discussion on this is provided in Section 2.

collateral constraints that are governed by real estate values into a monetary business cycle model.

Below, we delineate the methodology and present the data. We then discuss the results.

2.1. Methodology and data

We rely on the standard version of the turning point algorithm – extensively used in the business cycle literature – to analyse financial and business cycles (see [Bry and Boschan \(1971\)](#) and [Harding and Pagan \(2002\)](#)). In contrast to alternative methods, such as spectral analysis, unobserved component models or wavelet techniques, this approach does not rely on assumptions about the secular component. Furthermore, it allows us to contrast the degree of asymmetry inherent in financial and business cycles as it classifies cycles into expansionary and recessionary phases.

The methodology consists of two steps. First, turning points are extracted for financial cycle indicators and business cycle indicators. Second, using the turning points of the financial cycle indicators, we construct a reference financial cycle and date its turning points, as suggested in the business cycle context by [Harding and Pagan \(2006\)](#) and [Canova and Schlaepfer \(2015\)](#). Both steps are set out in detail below.

The **classical turning point algorithm** is as follows: Let x_t be a candidate peak of the stochastic process $x_t \in \{x_1, \dots, x_T\}$ (e.g., of log credit) if $x_t \in \max\{x_{t-2}; x_{t-1}; x_t; x_{t+1}; x_{t+2}\}$ and candidate trough if $x_t \in \min\{x_{t-2}; x_{t-1}; x_t; x_{t+1}; x_{t+2}\}$. A candidate is confirmed if the following six standard conditions are met:

- Peaks and troughs must alternate. If there is a violation, e.g., two troughs occur in sequence, the higher of the two is eliminated; if there are two peaks, the lower peak is erased.
- A peak must be higher than the previous trough and a trough must be lower than the previous peak.
- Five quarters is the minimum length of a cycle, e.g., peak to peak.
- Two quarters is the minimum phase of a cycle, where phase is peak to trough or trough to peak.
- Peaks or troughs at the first or last two quarters are excluded.
- Peaks at the beginning or the end are excluded if they are lower than the initial or last values. Similarly, troughs at the beginning or end are excluded if they are higher than the initial or last values.

We construct a **reference cycle** from two indicators in the following way: Let $x_t(i)$ be an indicator capturing the financial cycle, where, in our case, $i \in \{1, 2\}$. Given the set of turning points of the individual series, we calculate at each time period t the distance in quarters to the nearest peak and create a new time series, $m_t^p(i)$. Similarly, we derive an indicator at each time period measuring the distance to the nearest trough, say $m_t^t(i)$. Over all i , we aggregate the new measures, $m_t^p(i)$ and $m_t^t(i)$, on the basis of their median and point where the two aggregates reach their minimum. Low values indicate that the two measures are close to a common peak or, respectively, a common trough. Thus, local minima of the two measures are selected as candidate peaks and troughs of the reference cycle. Subsequently, the same censoring rules used to determine the turning points on the individual series are applied.

We summarise the constructed cycles using the average duration of expansionary phases (trough-to-peak or TP) and contractionary phases (peak-to-trough or PT), as well as the number of phases for each country and cycle. To analyse the relation of financial and business cycles over time, we use a measure of concordance – i.e. whether both cycles are in the same or a different phase, where phase is defined as either TP or PT. Two series are perfectly pro-cyclical (counter-cyclical) if the concordance index is equal to unity (zero).

Table 1

Financial variables exhibit higher *volatility* than business cycle variables. (Percent).

	Credit growth	House price growth	GDP growth
Mean	1.04	0.41	0.58
Std.dev.	1.66	2.28	1.03
Skew.	0.06	0.04	−0.23
Min	−9.36	−20.19	−7.11
Max	10.61	17.23	6.01
AT	1.21 (1.16)	0.67 (3.95)	0.59 (0.98)
BE	1.03 (1.62)	0.49 (1.76)	0.54 (0.72)
DE	0.67 (0.80)	−0.01 (0.82)	0.50 (0.99)
DK	0.83 (1.40)	0.32 (2.73)	0.43 (1.09)
ES	0.96 (1.78)	0.48 (2.79)	0.62 (0.80)
FI	1.05 (1.80)	0.32 (2.56)	0.59 (1.46)
FR	0.91 (0.99)	0.44 (1.42)	0.53 (0.56)
IE	1.82 (2.74)	0.59 (3.02)	1.10 (1.44)
IT	0.78 (1.47)	0.40 (3.28)	0.42 (0.87)
NL	1.31 (1.40)	0.47 (2.47)	0.56 (1.15)
PT	0.96 (2.41)	−0.10 (1.38)	0.61 (1.22)
SE	0.87 (1.68)	0.50 (1.94)	0.53 (1.16)
UK	1.16 (2.32)	0.68 (2.85)	0.56 (0.94)

Notes: The first three rows refer to the average across country statistics. “Std.dev.” is the standard deviation and “Skew.” the skewness. “Min” and “Max” indicate the minimum and maximum values across countries. The lower panel reports the mean and standard deviation (in brackets) for each of the 13 countries individually.

We apply this methodology to 13 EU countries (Austria, Belgium, Germany, Denmark, Spain, Finland, France, Ireland, Italy, Netherlands, Portugal, Sweden, and the United Kingdom from 1970Q1–2015Q2) – bringing together aggregate credit and house prices for financial cycles and GDP for business cycles. Data are seasonally adjusted, deflated and transformed into log levels.⁶ Growth rates of indicators are reported in the summary statistics in [Table 1](#), with variables analysed in first differenced form in order to account for non-stationary levels.

2.2. Results

2.2.1. Amplitude and duration of financial and business cycles

The amplitude and duration of financial cycles exceeds that of business cycles.

- The average amplitude of variation in financial variables – such as credit volumes and house prices – in EU countries has been around twice that of economic activity on the basis of historical volatilities (see [Table 1](#)).
- The average duration of financial cycles is 11.5 years and exceeds that of business cycles by more than 60% (see [Table 2](#)).

The duration of financial cycles is more heterogeneous across countries than that of business cycles. While the standard deviation of cycle duration across countries is only 2.6 for business cycles, it is 6.5 years for financial cycles. This illustrates country-specific differences. In particular, while for most countries the TP phase is longer, this is not the case for the TP phase of Germany, Italy, and Portugal.⁷

Estimates of the overall length of financial and business cycles are broadly in line with previous results in the literature.⁸ For instance, for Germany a duration of the financial cycle of 9.4 years

⁶ See [Schüler et al. \(2015\)](#) for more details.

⁷ [Chart 4](#) in the Appendix provides the exact time series dimension to the dating of PT and TP of the financial cycles and the business cycles for each country.

⁸ [Drehmann et al. \(2012\)](#) go beyond the algorithm as suggested by [Harding and Pagan \(2006\)](#) and [Canova and Schlaepfer \(2015\)](#) and impose that specific series must have turning points in some window around the turning points of the reference cycles. For comparability, we also impose this restriction in a robustness exercise, and results remain qualitatively the same.

Table 2

Financial cycles exhibit longer duration but lower asymmetry than business cycles. (Years).

	Financial cycle			Business cycle		
	Peak-to-trough	Trough-to-peak	Overall	Peak-to-trough	Trough-to-peak	Overall
AT	0.8 [1]	2.5 [1]	3.3 [2]	1.0 [8]	4.6 [7]	5.6 [15]
BE	4.9 [2]	16.1 [2]	21.0 [4]	0.9 [7]	5.4 [6]	6.3 [13]
DE	8.8 [4]	0.6 [3]	9.4 [7]	0.8 [9]	4.0 [8]	4.8 [17]
DK	1.6 [7]	4.4 [6]	6.0 [13]	1.0 [11]	2.8 [11]	3.8 [22]
ES	2.5 [4]	7.3 [3]	9.8 [7]	1.2 [6]	6.3 [5]	7.5 [11]
FI	4.5 [3]	6.4 [4]	10.9 [7]	1.2 [8]	4.3 [8]	5.5 [16]
FR	2.7 [4]	8.3 [3]	11.0 [7]	0.8 [4]	12.2 [3]	13.0 [7]
IE	4.6 [4]	5.6 [4]	10.3 [8]	1.3 [5]	7.8 [4]	9.1 [9]
IT	5.7 [3]	5.0 [3]	10.7 [6]	1.3 [10]	3.1 [9]	4.3 [19]
NL	3.5 [2]	25.3 [1]	28.8 [3]	1.0 [7]	5.5 [6]	6.5 [13]
PT	6.9 [2]	3.0 [2]	9.9 [4]	1.1 [8]	4.3 [7]	5.4 [15]
SE	1.8 [4]	8.2 [4]	10.0 [8]	1.3 [5]	8.9 [4]	10.2 [9]
UK	2.7 [5]	5.7 [5]	8.4 [10]	1.2 [5]	7.5 [4]	8.7 [9]
Mean	3.9	7.6	11.5	1.1	5.9	7.0
Std	2.3	6.5	6.5	0.2	2.6	2.6

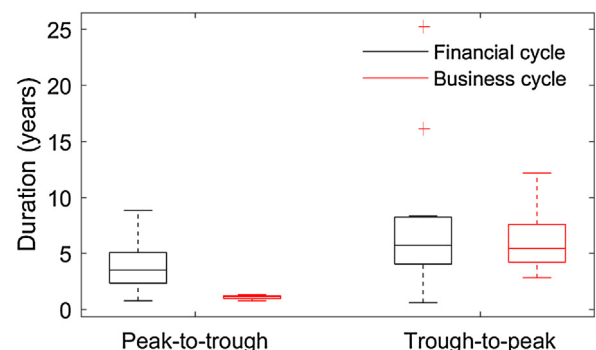
Notes: Mean duration in years is reported and the number of events in square brackets. Overall refers to the sum of peak-to-trough and trough-to-peak. “Std” refers to the standard deviation around the mean estimate of length across countries.

compares with results of [Schüler et al. \(2017\)](#) of 7.4 years, [Galati et al. \(2016\)](#) of 10.5 years, and [Rünstler and Vlekke \(2016\)](#) of 11.6 or 9.3 years, depending on the choice of model. Stronger differences emerge for Italy, where a financial cycle length of 10.7 years compares with 15.9 years in [Schüler et al. \(2017\)](#), 13.6 years in [Galati et al. \(2016\)](#), and for [Rünstler and Vlekke \(2016\)](#) 19.2 or 15.6 years, again depending on the choice of model. With respect to business cycles, our reported duration of contractions and upturns is similar to that in other studies (see [Claessens et al. \(2012\)](#) for contractions and [Hiebert et al. \(2014\)](#) for both downturns and upturns). Overall business cycle duration, which on average is 7 years, is also broadly in line with studies using other methodologies. For instance, using spectral analysis [Schüler et al. \(2017\)](#) find an overall duration of 6.7 years across G7 countries when excluding Japan. Using unobserved components models, [Rünstler and Vlekke \(2016\)](#) find an average duration of around 8.2 years for the univariate model, i.e., across France, Germany, Italy, United Kingdom, United States, and Spain.

2.2.2. Asymmetry of financial and business cycles

Financial cycles exhibit greater symmetry than business cycles.

- Already at the indicator level, real GDP growth is left-tailed across countries (negative skewness; see [Table 1](#)), while credit and house price growth tends to be symmetric (marginally positive skewness).
- Turning point analysis suggests that while some asymmetry is inherent in financial cycles – with expansions (7.6 years) tending to last longer than contractions (3.9 years) – the asymmetry in business cycles is stronger. For business cycles, contractions last on average only around 1.1 years, while expansions last around 5.9 years (see [Table 2](#) and [Chart 1](#)).

**Chart 1.** Financial cycles are more symmetric than business cycles.

Notes: Boxplots show distribution of duration (in years) of peak-to-trough and trough-to-peak phases across countries (see [Table 2](#)).

This finding of a short duration of contractionary phases of business cycles is in line with existing findings for several advanced economies. In particular, an extracted euro area reference business cycle has been shown to have the exact same asymmetry, i.e., 1.2 vs. 6.1 years in [Canova and Schlaepfer \(2015\)](#).⁹ The longer duration of financial cycle contractions than their business cycle counterparts illustrates the close connection of financial cycles to the concept of leverage cycles (see [Geanakoplos \(2010\)](#)), where deleveraging is typically time-consuming. This could also indicate that previous stabilization policies have been less focussed on the attenuation of financial cycles than on the attenuation of business cycles.

⁹ See also similar findings in [Hiebert et al. \(2014\)](#) for euro area countries, the NBER's Business Cycle Dating Committee for the US, or [Claessens et al. \(2012\)](#) for a large set of countries.

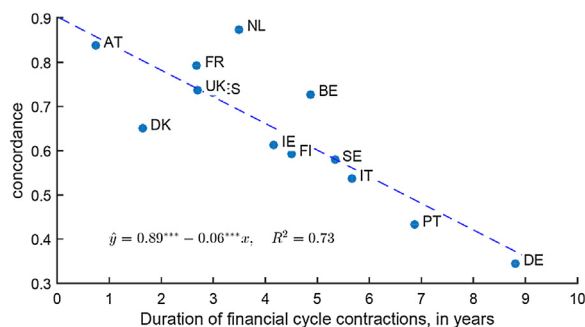


Chart 2. The longer the financial cycle contractions the lower the concordance with business cycles.

Notes: The blue dashed line represents an OLS fit and the below equation their estimates indicating their significance level at the 1% level and the respective R^2 measure.

2.2.3. Concordance of financial and business cycles

- Financial and business cycles share a mildly procyclical relationship (Table 3), being in phase two-thirds of the time over the last 45 years.¹⁰

This concordance, however, is quite heterogeneous across countries, and appears highest in the Netherlands (concordant 87% of the time) and weakest in Germany (concordant only 34% of the time).

The degree to which business and financial cycles synchronise at the country level bears a close relation to the duration of financial cycle contractions – where concordance with business cycles is all the higher, the shorter the average financial contractions are (see Chart 2).¹¹ This result is to some degree driven by the generally short contractionary phases of business cycles, where business cycles were largely in an expansionary phase for most of the countries over the past 45 years. In the extreme cases of Germany and Portugal, financial cycles were contracting for most of the considered sample (see Chart 4 in Appendix), and, thus, have the lowest concordance with their business cycle counterparts.

While there has been substantial variation of concordance of country cycles through time, concordance was very high preceding only three out of five CEPR dated recessions, with both cycles in expansionary phases (see Chart 3).¹² In general, however, the synchronisation of cycles tends to diverge during CEPR recession periods. Financial and business cycle synchronisation remained relatively low between 1980 and 1995. The period from 1995 until the outbreak of the global financial crises was the longest period in which financial and business cycles remained in an expansionary phase across countries; except for a downturn of some country's business cycles during the burst of the dot.com bubble (see Chart 4 in Appendix). Furthermore, euro area recessions since the 1990's have tended to be associated with the slight majority of countries in simultaneous financial and business cycle downturns. Clearly, simultaneous downturns are rare due to short periods of business cycle contractions. Overall, results indicate that financial cycles may remain in upturns during CEPR recession dates, indicating one dimension of divergence which is not driven by systematic lead-

¹⁰ This reading is halfway between 1 (which suggests perfect procyclicality, and zero perfectly countercyclicality), and a benchmark 0.5, which would represent a random relation between cycles.

¹¹ Further results are available upon request from the authors. We also experimented with the duration of expansionary phases (result: longer expansion leads to higher concordance) or the overall length of financial cycles relative to the length of business cycles (result: larger differences lead to lower concordance, with the exceptions of the Netherlands and Belgium). However, the highest explanatory power is achieved for the duration of contractionary phases. This is why we focus on this result.

¹² The starting dates of the three recessions are 1974Q4, 1980Q1, 2008Q1

lag relation of financial and business cycles, as can be inferred from Chart 4 in the Appendix.

The above results are broadly consistent with those of other studies. Our findings point out that shared phases of financial and business cycles appear to be largest over the medium term (more than 8 years). Rünstler and Vlekke (2016) find for a sample of five EU countries as well as the United States that cycles in financial variables (in credit volumes and house prices) and cycles in real economic activity are closely linked over the medium term but exhibit important divergences shorter business cycle frequencies. Similarly, Claessens et al. (2012) report that turning points in financial variables coincide with major booms and busts in economic activity, but that economic activity is subject to additional short-run fluctuations.

3. Economic mechanisms underlying different cycles: what can we learn from the theoretical literature?

While several of the stylized facts reported in Section 2 have been reported in the literature, relatively few efforts have been made to map such statistical findings on to candidate underlying economic mechanisms. Our results suggest that a successful candidate theoretical explanation would have to account for the limited concordance of financial and business cycles, by reproducing the three stylized facts identified in the preceding empirical section of this paper, namely: (i) the higher *volatility* of financial cycles, (ii) their higher *persistence*, and (iii) the *difference in symmetry* across the two cycles.

An initial complication is explaining a **higher amplitude of financial cycles**, stemming from the prospect for disconnect between asset prices and fundamentals. The success of DSGE models is due in large part to the ability of this class of models to reproduce the dynamics of business cycles. However, developing models that are able to simultaneously explain the dynamics of asset prices remains a formidable challenge (see, for example, Shiller (1981), and Mehra and Prescott (1985)). The main problem of standard DSGE models is their inability to account for the fact that some assets are riskier than others. While in the data the average difference between the return on a broad equity index and the short-term return on a risk-free government bond rate is about 6%, generating an equity premium of even 0.1% presents difficulties for the vast majority of business cycle models. A direct consequence of this inability to account for differences in risk characteristics is that the generation of DSGE models developed before the crisis typically fails to provide an explanation for the high volatility of asset prices, such as house prices, documented in Section 2.

A second challenge involves reproducing the **high persistence** of financial cycles relative to business cycles. This problem stems from the fact that the propagation mechanism of standard DSGE models is not sufficiently powerful to propagate and amplify the effects of exogenous shocks. As demonstrated by some key contributions to the business cycle literature, one potential solution to this problem is to introduce labour market externalities (e.g., Chang et al. (2002)). But since financial cycles are more persistent than business cycles, the puzzle reappears once financial and business cycle variables are studied in tandem. In such frameworks, the challenge is not only to generate fluctuations in output that are sufficiently persistent but also to reproduce the fact that financial cycles typically have a longer duration than business cycles.

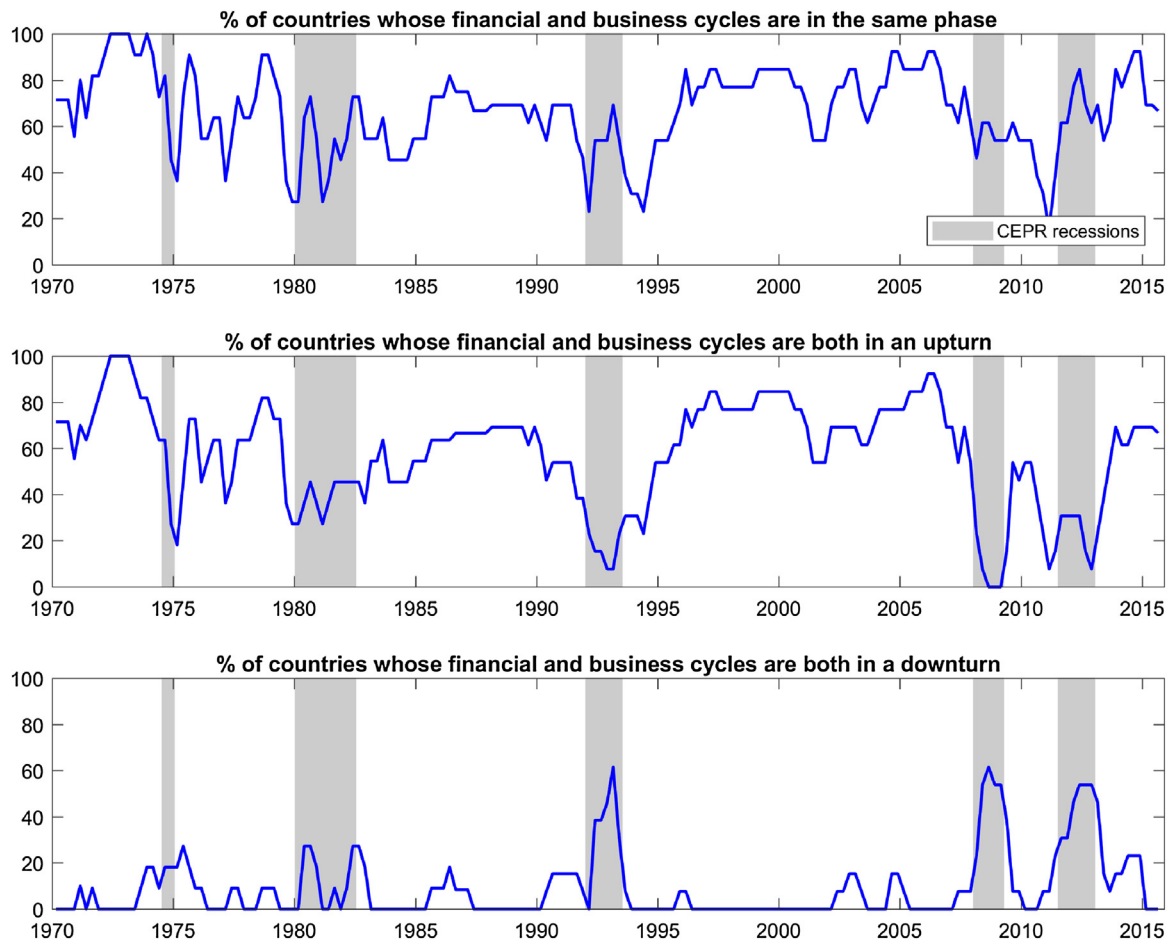
A third challenge is accounting for the **difference in symmetry between financial and business cycles**. Since business cycles are more asymmetric than financial cycles, it is necessary to recognize the non-linearity that is inherent to macroeconomic models. Accordingly, DSGE models that aim to capture asymmetries need to be solved using high-order approximation methods. This has the

Table 3

Concordance of financial and business cycles differs considerably across countries.

NL	AT	FR	SE	UK	ES	BE	IE	DK	FI	IT	PT	DE	Average
0.87	0.84	0.79	0.77	0.74	0.72	0.71	0.66	0.65	0.59	0.54	0.43	0.34	0.67

Notes: Concordance measures the fraction of the time both the financial and the business cycle are in the same phase. The series are perfectly pro-cyclical (counter-cyclical) if the concordance index is equal to unity (zero).

**Chart 3.** Large percentage of countries with both financial and business cycle in upturn ahead of the global financial crisis time.

potential to reproduce the skewness in distribution that is sometimes observed in the data.

3.1. A selective literature review

Several candidate economic mechanisms can account for some of above-mentioned challenges – namely asset price disconnect inherent to distinct financial cycles, high persistence of financial cycles, and asymmetry in business cycles. Five candidate mechanisms are reviewed: (i) debt overhang, (ii) financial shocks, (iii) learning and deviations from rational expectations, (iv) strategic complementarities and (v) financial structures and credit booms.

3.1.1. Debt overhang

In a recent paper, Kalemli-Özcan et al. (2015) find that the sluggish dynamics of private investment observed in Europe since the onset of the crisis is caused by a significant debt overhang effect. Overall, the results documented in this study suggest that debt overhang has been holding back private-sector investment and that this effect is linked to changes in sovereign risk. Together with the high level of debt recorded in many countries, the creditless recov-

ery observed in some Eurozone economies since the end of the subprime crisis seems consistent with this mechanism. As pointed out by Andrés et al. (2014), deleveraging played a central role in some euro area economies, such as Spain, and credit typically recovers much more slowly than output when the economy enters a deleveraging regime. This is also consistent with the empirical evidence documented by Claessens et al. (2009) who showed that recessions associated with credit crunches and house price busts tend to be deeper and longer than other recessions. Episodes of deleveraging are typical examples during which measures of the financial cycle, such as credit and house prices, recover more slowly than output, and could therefore play a role in explaining the longer duration of financial cycle downturns uncovered in our empirical section.

As shown in Myers (1977), the distortion caused by debt overhang is due to the fact that firms with high levels of debt have a high probability of defaulting. Since debt is senior to equity, any proceeds from new investment will serve to repay debt holders and will therefore flow mostly to the firms' creditors, thereby reducing the incentive of equity owners, who are the ones who make investment decisions, to pursue costly new projects.

In standard DSGE models, the level of indebtedness has no effect on the firm investment decision because these models typically abstract from default risk. In a first step to fill this gap, a recent paper by [Occhino and Pescatori \(2015\)](#) studies this question in a model in which the interaction between the financial and business cycles is affected by a debt overhang distortion. The main finding which is relevant for our purpose is that this distortion contributes to amplifying and propagating the effects of macroeconomic shocks. In a model in which firms issue long-term debt, [Gomes et al. \(2016\)](#) show that a debt overhang that persists for many periods can be created by low inflation. In this environment, long-term debt and default provide a channel through which leverage can cause financially driven recessions.

As shown by [Lamont \(1995\)](#), one unintended consequence of corporate debt overhang is that it can create multiple equilibria. Debt overhang deters new investment because the benefit of new investment will go the existing creditors, not the investors if the firm goes bankrupt. As a result, investors only undertake projects with a net present value that is sufficiently high when the firm is in a debt overhang situation. Depending on expectations about the net present value of a project, which, in turn, depends on macroeconomic conditions, high levels of debt can thus create a channel of interaction between business and financial cycles in the sense that it makes the economy more vulnerable to changes in expectations.

3.1.2. The role of financial shocks

One main lesson from the 2007–2009 global crisis is that financial shocks play a more important role than what is typically predicted by models developed during the Great Moderation period. [Jermann and Quadrini \(2012\)](#), for instance, find that financial shocks contribute to almost half of the volatility of output and about 30 percent to the volatility of hours worked. In their model, the price mark-up shock explains about one quarter of output fluctuations and more than half of debt repurchases. [Gertler and Karadi \(2011\)](#) obtain persistence by combining capital quality shocks that simultaneously affect the production function and the accumulation of capital with financial frictions. The findings reported by [Christiano et al. \(2014\)](#) suggest that risk shocks are the main drivers of output fluctuations. In their framework, risk shocks play a particularly important role for financial variables. In [Jaccard \(2018\)](#), financial shocks that destroy the stock of safe assets used as collateral by consumers generate slow recoveries, a stock market crash as well as a flight-to-safety effect.

Overall, the results documented in the literature on financial shocks that has emerged since the crisis suggest that interacting this type of shock with financial frictions could strengthen the propagation mechanism of DSGE models, thereby improving their ability to account for the higher volatility and persistence of financial cycle variables observed in the data.

3.1.3. Learning and deviation from rational expectations

A strand of the literature suggests that departure from rational expectations could provide an explanation for some of the facts documented in Section 2. Most of the DSGE literature is built on the assumption that agents form rational expectations (e.g., [Muth \(1961\)](#); [Lucas \(1976\)](#)). The main motivation for introducing deviations from rational expectations is the lack of persistence that is typically obtained in DSGE models. A simple way of addressing this problem is to introduce backward-looking components into the model by assuming that expectations are myopic, in the sense that they depend on past observations.

Recent contributions to the literature consider departure from the rational expectations paradigm by considering more realistic forms of learning schemes that differentiate between different degrees of rationality. In particular, [Adam and Marcet \(2011\)](#) introduce a distinction between internal and external rationality, where

internal rationality refers to the requirement that agents make fully optimal decisions given a well-defined system of subjective probability beliefs. External rationality refers to the stronger requirement that agents have a complete understanding of the system in which they evolve and therefore that agents' subjective probability beliefs equal the objective probability density of the stochastic variables that emerges in equilibrium. While the mechanism developed by [Adam and Marcet \(2011\)](#) maintains the assumption of internal rationality, deviations from the pure rational expectation hypothesis are introduced by relaxing the stronger assumption of external rationality.

In the context of an endowment economy model, introducing such deviations from the rational expectation paradigm significantly improves the ability of a simple model to account for some asset pricing puzzles. In particular, once augmented with this mechanism, the endowment economy model becomes quantitatively consistent with the observed volatility of stock prices (e.g., [Adam et al. \(2017\)](#)).

Other recent contributions to the adaptive learning literature include the paper by [Lansing \(2010\)](#) who shows that a near rational solution helps an otherwise standard endowment economy model to reproduce some asset pricing facts. Using a similar strategy, [Gelain et al. \(2013\)](#) study a model in which deviations from rational expectations increase the persistence of house prices and household debt. Other applications of learning include the work by [Hennessy and Radnaev \(2018\)](#) who develop a DSGE model in which firms learn about disaster risk. During periods of low volatility, this learning mechanism reduces the perceived probability of tail risk, which, in turn, leads to excessive leverage.

3.1.4. Strategic complementarities

As summarized by [De Nicolo et al. \(2012\)](#), one reason why agents choose to correlate their risk is because of strategic complementarities, in the sense that the payoff from a certain strategy increases with the number of participants who choose to adopt this particular strategy. A classic example is provided by [Rajan \(1994\)](#) who studies the potential causes of the banking crisis that hit New England in the early 1990s. In particular, this study assesses the factors that could rationalize the observed behaviour of banks whose managers sometimes choose to fund projects with a negative net present value and tend to underreport losses in periods of economic downturn. The mechanism studied in this paper builds on the assumption that information asymmetry creates strategic complementarities between bank credit policies, making a bank's reputation less sensitive to poor earnings when other banks admit poor earnings. Using global games techniques, [Aikman et al. \(2015\)](#) show that sources of strategic complementarity between banks, such as reputational concerns, for instance, open the door for small changes in fundamentals to generate large swings in credit relative to real activity.

3.1.5. Financial structures and credit booms

Differences in market structures could be a factor responsible for the divergence between financial and business cycle observed in the data. They could also explain the heterogeneity in the degree of synchronization across countries. The findings of [Rünstler and Vlekke \(2016\)](#) suggest that cross-country differences in financial cycles can be attributed to structural characteristics of the housing markets (see also [Huber \(2016\)](#)). The results documented by [De Fiore and Uhlig \(2015\)](#) also suggest that structural features of the economy such as the extent to which firms have access to market-based financing influence the dynamics of both business and financial cycle variables.

Recent empirical literature has uncovered a link between periods of rapid credit growth and financial crises ([Drehmann et al. \(2012\)](#); [Jordà et al. \(2015\)](#); [Aikman et al. \(2015\)](#)). Overall, the empirical evidence documented by this strand of the literature shows that

credit and leverage play an active role in influencing business cycle dynamics. Boissay et al. (2016) develop a model in which banking crises are rare events that break out in the midst of credit intensive booms. Banking crises result from the pro-cyclicality of bank balance sheets, whereby lower rates aggravate agency problems in the interbank markets, which, in turn, lead to a reduction in market funding. Since in this model the household's stock of asset is pre-determined, this mechanism can generate financial cycles that are longer than business cycles.

In Mendoza (2010), sudden stops are studied in a model in which an occasionally binding collateral constraint amplifies the effects of macroeconomic shocks. In this environment, sudden stops are caused by credit frictions. This mechanism can generate large recessions as well as asymmetric fluctuations. Guerrieri and Iacoviello (2017) estimate a DSGE model with a housing sector in which the effects of adverse shocks are amplified by occasionally binding collateral constraints. Relative to the model studied in Iacoviello (2005), the novel solution techniques developed in their study can be used to estimate a model in which business cycle asymmetries are caused by occasionally binding constraints.

4. Conclusions

This study finds that financial cycles and business cycles are distinct in three ways. First, financial cycles are larger than their business cycle counterparts. Second, they are longer, and third, financial cycles are more symmetric – exhibiting, in particular, more protracted downturns than business cycles. Mostly owing to the difference in symmetry, we find that financial and business cycles are imperfectly synchronized. Ultimately, the longer duration of financial cycle downturns represents a key dimension of divergence.

Several candidate economic mechanisms may underlie these diverging features – namely, an intermittent tendency for disconnect between credit and asset prices from their fundamentals, as well as a relatively higher degree of persistence and symmetry of financial cycles. These mechanisms range from the powerful lasting effects of debt overhang and potential for multiple equilibria, the accelerator type properties of financial shocks, scope for expectational errors (such as learning and deviations from rational expectations), and the role for externalities and associated market failure (for instance, stemming from strategic complementarities).

Taken together, these findings suggest scope for carefully calibrated policies attuned to divergent cycles at their root economic causes. While a debate on the merits of having monetary policy lean against the wind is still ongoing,¹³ our results suggest that using monetary policy as the first line of defence against financial instability may lead to conflicts between price and financial stability objectives – at least over shorter horizons. As demonstrated by our results, the first main issue is the fact that financial and business cycles are only imperfectly synchronized. The second main risk is that a “leaning against the wind” policy could increase the volatility of business cycle fluctuations while only marginally reducing the probability of financial crises. Adding a separate tool, such as a macroprudential instrument or fiscal policy, may therefore be necessary to address the root cause of financial instability.¹⁴

¹³ On the one hand, several studies suggest monetary policy has only a moderate impact on the probability of financial crises (e.g., Svensson (2017)). Moreover, Dell’Ariccia et al. (2017), using U.S. data, find that monetary policy has a statistically significant effect but only a small economic effect on risk-taking by financial intermediaries. On the other hand, the BIS has strongly argued that monetary policy should target financial as well as business cycles (e.g. BIS (2017)).

¹⁴ See, for instance, Alpanda and Zubairy (2017) for a cost-benefit analysis of the effects of fiscal policy and macroprudential regulation in reducing household indebted-

Appendix A.

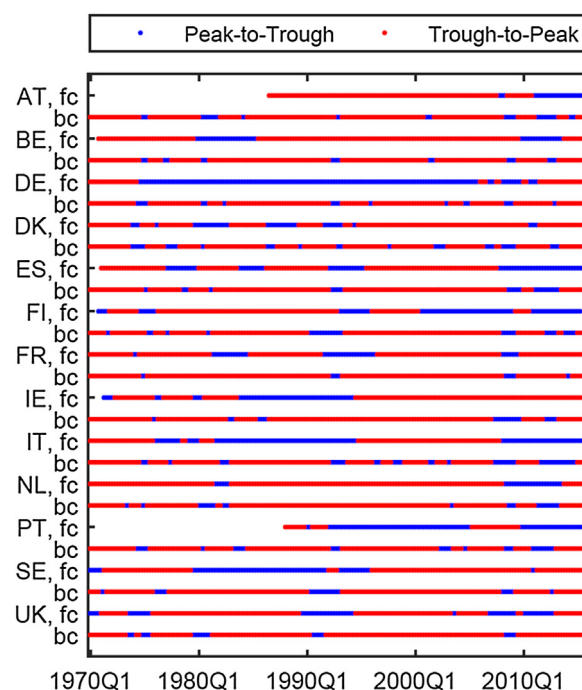


Chart 4. Distribution of trough-to-peak and peak-to-trough of financial and business cycles over.

Notes: “fc” refers to financial cycles and “bc” to business cycle.

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